

Deviations of random matrices and geometric consequences

This chapter is devoted to a remarkably useful uniform deviation inequality for random matrices. Given an $m \times n$ random matrix A , our goal is to show that with high probability, the approximate equality

$$\|Ax\|_2 \approx \mathbb{E} \|Ax\|_2 \quad (9.1)$$

holds *simultaneously for many vectors* $x \in \mathbb{R}^n$. To quantify how many, we may choose an arbitrary subset $T \subset \mathbb{R}^n$ and ask whether (9.1) holds simultaneously for all $x \in T$. The answer turns out to be remarkably simple: with high probability, we have

$$\|Ax\|_2 = \mathbb{E} \|Ax\|_2 + O(\gamma(T)) \quad \text{for all } x \in T. \quad (9.2)$$

Recall that $\gamma(T)$ is the Gaussian complexity of T , which is a cousin of Gaussian width we introduced in Section 7.6.2. In Section 9.1, we deduce the uniform deviation inequality (9.2) from Talagrand's comparison inequality.

The uniform matrix deviation inequality has many consequences. Some of them are results we proved earlier by different methods: in Section 9.2–9.3 we quickly deduce two-sided bounds on random matrices, bounds on random projections of geometric sets, guarantees for covariance estimation for lower-dimensional distributions, Johnson-Lindenstrauss Lemma and its generalization for infinite sets. New consequences will be proved starting from Section 9.4, where we deduce two classical results about geometric sets in high dimensions: the M^* bound and the Escape theorem. Applications to sparse signal recovery will follow in Chapter 10.

9.1 Matrix deviation inequality

The following theorem is the main result of this chapter.

Theorem 9.1.1 (Matrix deviation inequality). *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then for any subset $T \subset \mathbb{R}^n$, we have*

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m} \|x\|_2 \right| \leq CK^2 \gamma(T).$$

Here $\gamma(T)$ is the Gaussian complexity introduced in Section 7.6.2, and $K = \max_i \|A_i\|_{\psi_2}$.

Before we proceed to the proof of this theorem, let us pause to check that $\mathbb{E} \|Ax\|_2 \approx \sqrt{m} \|x\|_2$, so Theorem 9.1.1 indeed yields (9.2).

Exercise 9.1.2 (Deviation around expectation).  Deduce from Theorem 9.1.1 that

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2 - \mathbb{E} \|Ax\|_2 \right| \leq CK^2 \gamma(T).$$

Hint: Bound the difference between $\mathbb{E} \|Ax\|_2$ and $\sqrt{m} \|x\|_2$ using concentration of norm (Theorem 3.1.1).

We will deduce Theorem 9.1.1 from Talagrand's comparison inequality (Corollary 8.6.3, and more specifically Exercise 8.6.4). To apply the comparison inequality, all we have to check is that the random process

$$X_x := \|Ax\|_2 - \sqrt{m} \|x\|_2$$

indexed by $x \in \mathbb{R}^n$ has sub-gaussian increments. Let us state this.

Theorem 9.1.3 (Sub-gaussian increments). *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random process*

$$X_x := \|Ax\|_2 - \sqrt{m} \|x\|_2$$

has sub-gaussian increments, namely

$$\|X_x - X_y\|_{\psi_2} \leq CK^2 \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n. \quad (9.3)$$

Here $K = \max_i \|A_i\|_{\psi_2}$.

Proof of matrix deviation inequality (Theorem 9.1.1) By Theorem 9.1.3 and Talagrand's comparison inequality in the form of Exercise 8.6.4, we get

$$\mathbb{E} \sup_{x \in T} |X_x| \leq CK^2 \gamma(T)$$

as announced. □

It remains to prove Theorem 9.1.3. Although the proof is a bit longer than most of the arguments in this book, we make it easier by working out simpler, partial cases first and gradually moving toward full generality. We develop this argument in the next few subsections.

9.1.1 Theorem 9.1.3 for unit vector x and zero vector y

Assume that

$$\|x\|_2 = 1 \quad \text{and} \quad y = 0.$$

In this case, the inequality in (9.3) we want to prove becomes

$$\left\| \|Ax\|_2 - \sqrt{m} \right\|_{\psi_2} \leq CK^2. \quad (9.4)$$

Note that Ax is a random vector in $\mathbb{R}^m$ with independent, sub-gaussian coordinates $\langle A_i, x \rangle$, which satisfy $\mathbb{E} \langle A_i, x \rangle^2 = 1$ by isotropy. Then the Concentration of Norm (Theorem 3.1.1) yields (9.4).

9.1.2 Theorem 9.1.3 for unit vectors x, y and for the squared process

Assume now that

$$\|x\|_2 = \|y\|_2 = 1.$$

In this case, the inequality in (9.3) we want to prove becomes

$$\left\| \|Ax\|_2 - \|Ay\|_2 \right\|_{\psi_2} \leq CK^2 \|x - y\|_2. \quad (9.5)$$

We first prove a version of this inequality for the *squared* Euclidean norms, which are more convenient to handle. Let us guess what form such inequality should take. We have

$$\begin{aligned} \|Ax\|_2^2 - \|Ay\|_2^2 &= \left(\|Ax\|_2 + \|Ay\|_2 \right) \cdot \left(\|Ax\|_2 - \|Ay\|_2 \right) \\ &\lesssim \sqrt{m} \cdot \|x - y\|_2. \end{aligned} \quad (9.6)$$

The last bound should hold with high probability because the typical magnitude of $\|Ax\|_2$ and $\|Ay\|_2$ is $\sqrt{m}$ by (9.4) and since we expect (9.5) to hold.

Now that we guessed the inequality (9.6) for the squared process, let us prove it. We are looking to bound the random variable

$$Z := \frac{\|Ax\|_2^2 - \|Ay\|_2^2}{\|x - y\|_2} = \frac{\langle A(x - y), A(x + y) \rangle}{\|x - y\|_2} = \langle Au, Av \rangle \quad (9.7)$$

where

$$u := \frac{x - y}{\|x - y\|_2} \quad \text{and} \quad v := x + y.$$

The desired bound is

$$|Z| \lesssim \sqrt{m} \quad \text{with high probability.}$$

The coordinates of the vectors Au and Av are $\langle A_i, u \rangle$ and $\langle A_i, v \rangle$. So we can represent Z as a sum of independent random variables

$$Z = \sum_{i=1}^m \langle A_i, u \rangle \langle A_i, v \rangle,$$

Lemma 9.1.4. *The random variables $\langle A_i, u \rangle \langle A_i, v \rangle$ are independent, mean zero, and sub-exponential; more precisely,*

$$\left\| \langle A_i, u \rangle \langle A_i, v \rangle \right\|_{\psi_1} \leq 2K^2.$$

Proof Independence follows from the construction, but the mean zero property is less obvious. Although both $\langle A_i, u \rangle$ and $\langle A_i, v \rangle$ do have zero means, these variables are not necessarily independent from each other. Still, we can check that they are uncorrelated. Indeed,

$$\mathbb{E} \langle A_i, x - y \rangle \langle A_i, x + y \rangle = \mathbb{E} \left[\langle A_i, x \rangle^2 - \langle A_i, y \rangle^2 \right] = 1 - 1 = 0$$

by isotropy. By definition of u and v , this implies that $\mathbb{E} \langle A_i, u \rangle \langle A_i, v \rangle = 0$.

To finish the proof, recall from Lemma 2.7.7 that the product of two sub-gaussian random variables is sub-exponential. So we get

$$\begin{aligned} \left\| \langle A_i, u \rangle \langle A_i, v \rangle \right\|_{\psi_1} &\leq \left\| \langle A_i, u \rangle \right\|_{\psi_2} \cdot \left\| \langle A_i, v \rangle \right\|_{\psi_2} \\ &\leq K \|u\|_2 \cdot K \|v\|_2 \quad (\text{by sub-gaussian assumption}) \\ &\leq 2K^2 \end{aligned}$$

where in the last step we used that $\|u\|_2 = 1$ and $\|v\|_2 \leq \|x\|_2 + \|y\|_2 \leq 2$. $\square$

To bound Z , we can use Bernstein's inequality (Corollary 2.8.3); recall that it applies for a sum of independent, mean zero, sub-exponential random variables.

Exercise 9.1.5.  Apply Bernstein's inequality (Corollary 2.8.3) and simplify the bound. You should get

$$\mathbb{P} \left\{ |Z| \geq s\sqrt{m} \right\} \leq 2 \exp \left(- \frac{cs^2}{K^4} \right)$$

for any $0 \leq s \leq \sqrt{m}$.

Hint: In this range of s , the sub-gaussian tail will dominate in Bernstein's inequality. Do not forget to apply the inequality for $2K^2$ instead of K because of Lemma 9.1.4.

Recalling the definition of Z , we can see that we obtained the desired bound (9.6).

9.1.3 Theorem 9.1.3 for unit vectors x, y and for the original process

Next, we would like to remove the squares from $\|Ax\|_2^2$ and $\|Ay\|_2^2$ and deduce inequality (9.5) for unit vectors x and y . Let us state this goal again.

Lemma 9.1.6 (Unit y , original process). *Let $x, y \in S^{n-1}$. Then*

$$\left\| \|Ax\|_2 - \|Ay\|_2 \right\|_{\psi_2} \leq CK^2 \|x - y\|_2.$$

Proof Fix $s \geq 0$. By definition of the sub-gaussian norm,¹ the conclusion we want to prove is that

$$p(s) := \mathbb{P} \left\{ \frac{\left| \|Ax\|_2 - \|Ay\|_2 \right|}{\|x - y\|_2} \geq s \right\} \leq 4 \exp \left(- \frac{cs^2}{K^4} \right). \quad (9.8)$$

¹ Recall (2.14) and Remark 2.5.3.

We proceed differently for small and large s .

Case 1: $s \leq 2\sqrt{m}$. In this range, we use our results from the previous subsection. They are stated for the squared process though. So, to be able to apply those results, we multiply both sides of the inequality defining $p(s)$ by $\|Ax\|_2 + \|Ay\|_2$. With the same Z as we defined in (9.7), this gives

$$p(s) = \mathbb{P} \left\{ |Z| \geq s(\|Ax\|_2 + \|Ay\|_2) \right\} \leq \mathbb{P} \left\{ |Z| \geq s\|Ax\|_2 \right\}.$$

From (9.4) we know that $\|Ax\|_2 \approx \sqrt{m}$ with high probability. So it makes sense to break the probability that $|Z| \geq s\|Ax\|_2$ into two cases: one where $\|Ax\|_2 \geq \sqrt{m}/2$ and thus $|Z| \geq s\sqrt{m}/2$, and the other where $\|Ax\|_2 < \sqrt{m}/2$ (and there we do not care about Z). This leads to

$$p(s) \leq \mathbb{P} \left\{ |Z| \geq \frac{s\sqrt{m}}{2} \right\} + \mathbb{P} \left\{ \|Ax\|_2 < \frac{\sqrt{m}}{2} \right\} =: p_1(s) + p_2(s).$$

The result of Exercise 9.1.5 gives

$$p_1(s) \leq 2 \exp \left(-\frac{cs^2}{K^4} \right).$$

Further, the bound (9.4) and triangle inequality gives

$$p_2(s) \leq \mathbb{P} \left\{ \left| \|Ax\|_2 - \sqrt{m} \right| > \frac{\sqrt{m}}{2} \right\} \leq 2 \exp \left(-\frac{cs^2}{K^4} \right).$$

Summing the two probabilities, we conclude a desired bound

$$p(s) \leq 4 \exp \left(-\frac{cs^2}{K^4} \right).$$

Case 2: $s > 2\sqrt{m}$. Let us look again at the inequality (9.8) that defines $p(s)$, and slightly simplify it. By triangle inequality, we have

$$\left| \|Ax\|_2 - \|Ay\|_2 \right| \leq \|A(x - y)\|_2.$$

Thus

$$\begin{aligned} p(s) &\leq \mathbb{P} \left\{ \|Au\|_2 \geq s \right\} \quad (\text{where } u := \frac{x - y}{\|x - y\|_2} \text{ as before}) \\ &\leq \mathbb{P} \left\{ \|Au\|_2 - \sqrt{m} \geq s/2 \right\} \quad (\text{since } s > 2\sqrt{m}) \\ &\leq 2 \exp \left(-\frac{cs^2}{K^4} \right) \quad (\text{using (9.4) again}). \end{aligned}$$

Therefore, in both cases we obtained the desired estimate (9.8). This completes the proof of the lemma. $\square$

9.1.4 Theorem 9.1.3 in full generality

Finally, we are ready to prove (9.3) for arbitrary $x, y \in \mathbb{R}^n$. By scaling, we can assume without loss of generality that

$$\|x\|_2 = 1 \quad \text{and} \quad \|y\|_2 \geq 1. \quad (9.9)$$

(Why?) Consider the contraction of y onto the unit sphere, see Figure 9.1:

$$\bar{y} := \frac{y}{\|y\|_2} \quad (9.10)$$

Use triangle inequality to break the increment in two parts:

$$\|X_x - X_y\|_{\psi_2} \leq \|X_x - X_{\bar{y}}\|_{\psi_2} + \|X_{\bar{y}} - X_y\|_{\psi_2}.$$

Since x and $\bar{y}$ are unit vectors, Lemma 9.1.6 may be used to bound the first part. It gives

$$\|X_x - X_{\bar{y}}\|_{\psi_2} \leq CK^2 \|x - \bar{y}\|_2.$$

To bound the second part, note that $\bar{y}$ and y are collinear vectors, so

$$\|X_{\bar{y}} - X_y\|_{\psi_2} = \|\bar{y} - y\|_2 \cdot \|X_{\bar{y}}\|_{\psi_2}.$$

(Check this identity!) Now, since $\bar{y}$ is a unit vector, (9.4) gives

$$\|X_{\bar{y}}\|_{\psi_2} \leq CK^2.$$

Combining the two parts, we conclude that

$$\|X_x - X_y\|_{\psi_2} \leq CK^2 (\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2). \quad (9.11)$$

At this point we might get nervous: we need to bound the right hand side by $\|x - y\|_2$, but triangle inequality would give the reverse bound! Nevertheless, looking at Figure 9.1 we may suspect that in our case triangle inequality can be approximately reversed. The next exercise confirms this rigorously.

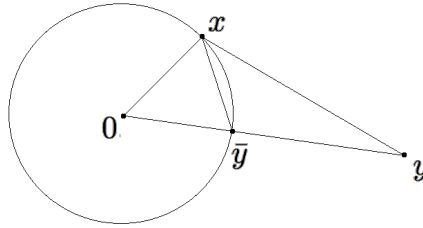


Figure 9.1 Exercise 9.1.7 shows that triangle inequality can be approximately reversed from these three vectors, and we have $\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2 \leq \sqrt{2}\|x - y\|_2$.

Exercise 9.1.7 (Reverse triangle inequality). Consider vectors $x, y, \bar{y} \in \mathbb{R}^n$ be satisfying (9.9) and (9.10). Show that

$$\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2 \leq \sqrt{2}\|x - y\|_2.$$

Using the result of this exercise, we deduce from (9.11) the desired bound

$$\|X_x - X_y\|_{\psi_2} \leq CK^2 \|x - y\|_2.$$

Theorem 9.1.3 is completely proved. $\square$

Now that we proved matrix deviation inequality (Theorem 9.1.1), we can complement it with the a high-probability version.

Exercise 9.1.8 (Matrix deviation inequality: tail bounds). $\clubsuit$ Show that, under the conditions of Theorem 9.1.1, we have the following. For any $u \geq 0$, the event

$$\sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m} \|x\|_2 \right| \leq CK^2 [w(T) + u \cdot \text{rad}(T)] \quad (9.12)$$

holds with probability at least $1 - 2\exp(-u^2)$. Here $\text{rad}(T)$ is the radius of T introduced in (8.47).

Hint: Use the high-probability version of Talagrand's comparison inequality from Exercise 8.6.5.

Exercise 9.1.9. $\clubsuit$ Argue that the right hand side of (9.12) can be further bounded by $CK^2 u \gamma(T)$ for $u \geq 1$. Conclude that the bound in Exercise 9.1.8 implies Theorem 9.1.1.

Exercise 9.1.10 (Deviation of squares). $\clubsuit\clubsuit$ Show that, under the conditions of Theorem 9.1.1, we have

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2^2 - m \|x\|_2^2 \right| \leq CK^4 \gamma(T)^2 + CK^2 \sqrt{m} \text{rad}(T) \gamma(T).$$

Hint: Reduce it to the original deviation inequality using the identity $a^2 - b^2 = (a - b)(a + b)$.

Exercise 9.1.11 (Deviation of random projections). $\clubsuit\clubsuit\clubsuit\clubsuit$ Prove a version of matrix deviation inequality (Theorem 9.1.1) for random projections. Let P be the orthogonal projection in $\mathbb{R}^n$ on an m -dimensional subspace uniformly distributed in the Grassmanian $G_{n,m}$. Show that for any subset $T \subset \mathbb{R}^n$, we have

$$\mathbb{E} \sup_{x \in T} \left| \|Px\|_2 - \sqrt{\frac{m}{n}} \|x\|_2 \right| \leq \frac{CK^2 \gamma(T)}{\sqrt{n}}.$$

9.2 Random matrices, random projections and covariance estimation

Matrix deviation inequality has a number of important consequences, some which we present in this and next section.

9.2.1 Two-sided bounds on random matrices

To get started, let us apply the matrix deviation inequality for the unit Euclidean sphere $T = S^{n-1}$. In this case, we recover the bounds on random matrices that we proved in Section 4.6.

Indeed, the radius and Gaussian width of $T = S^{n-1}$ satisfy

$$\text{rad}(T) = 1, \quad w(T) \leq \sqrt{n}.$$

(Recall (7.16).) Matrix deviation inequality in the form of Exercise 9.1.8 together with triangle inequality imply that the event

$$\sqrt{m} - CK^2(\sqrt{n} + u) \leq \|Ax\|_2 \leq \sqrt{m} + CK^2(\sqrt{n} + u) \quad \forall x \in S^{n-1}$$

holds with probability at least $1 - 2\exp(-u^2)$.

We can interpret this event as a two-sided bound on the extreme singular values of A (recall (4.5)):

$$\sqrt{m} - CK^2(\sqrt{n} + u) \leq s_n(A) \leq s_1(A) \leq \sqrt{m} + CK^2(\sqrt{n} + u).$$

Thus we recover the result we proved in Theorem 4.6.1.

9.2.2 Sizes of random projections of geometric sets

Another immediate application of matrix deviation inequality is the bound on random projections of geometric sets we gave in Section 7.7. In fact, matrix deviation inequality yields a sharper bound:

Proposition 9.2.1 (Sizes of random projections of sets). *Consider a bounded set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the scaled matrix*

$$P := \frac{1}{\sqrt{n}}A$$

(a “sub-gaussian projection”) satisfies

$$\mathbb{E} \text{diam}(PT) \leq \sqrt{\frac{m}{n}} \text{diam}(T) + CK^2 w_s(T).$$

Here $w_s(T)$ is the spherical width of T (recall Section 7.5.2) and $K = \max_i \|A_i\|_{\psi_2}$.

Proof Theorem 9.1.1 implies via triangle inequality that

$$\mathbb{E} \sup_{x \in T} \|Ax\|_2 \leq \sqrt{m} \sup_{x \in T} \|x\|_2 + CK^2 \gamma(T).$$

We can state this inequality in terms of radii of the sets AT and T as

$$\mathbb{E} \text{rad}(AT) \leq \sqrt{m} \text{rad}(T) + CK^2 \gamma(T).$$

Applying this bound for the difference set $T - T$ instead of T , we can write it as

$$\mathbb{E} \text{diam}(AT) \leq \sqrt{m} \text{diam}(T) + CK^2 w(T).$$

(Here we used (7.22) to pass from Gaussian complexity to Gaussian width.) Dividing both sides by $\sqrt{n}$ completes the proof. $\square$

Proposition 9.2.1 is more general and sharper than our older bounds on random projections (Exercise 7.7.3. Indeed, it states that the diameter scales by the exact factor $\sqrt{m/n}$ without an absolute constant in front of it.

Exercise 9.2.2 (Sizes of projections: high-probability bounds). ☹☹ Use the high-probability version of matrix deviation inequality (Exercise 9.1.8) to obtain a high-probability version of Proposition 9.2.1. Namely, show that for $\varepsilon > 0$, the bound

$$\text{diam}(PT) \leq (1 + \varepsilon) \sqrt{\frac{m}{n}} \text{diam}(T) + CK^2 w_s(T)$$

holds with probability at least $1 - \exp(-c\varepsilon^2 m/K^4)$.

Exercise 9.2.3. ☹☹☹ Deduce a version of Proposition 9.2.1 for the original model of P considered in Section 7.7, i.e. for a random projection P onto a random m -dimensional subspace $E \sim \text{Unif}(G_{n,m})$.

Hint: If $m \ll n$, the random matrix A in matrix deviation inequality is an approximate projection: this follows from Section 4.6.

9.2.3 Covariance estimation for lower-dimensional distributions

Let us revisit the covariance estimation problem, which we studied in Section 4.7 for sub-gaussian distributions and in Section 5.6 in full generality. We found that the covariance matrix Σ of an n -dimensional distribution can be estimated from $m = O(n)$ sample points for sub-gaussian distributions, and from $m = O(n \log n)$ sample points in full generality.

An even smaller sample can be sufficient for covariance estimation when the distribution is approximately low-dimensional, i.e. when $\Sigma^{1/2}$ has low stable rank,² which means that the distribution tends to concentrate near a small subspace in $\mathbb{R}^n$. We should expect to do well with $m = O(r)$, where r is the stable rank of $\Sigma^{1/2}$. We noted this only for the general case in Remark 5.6.3, up to a logarithmic oversampling. Now let us address the sub-gaussian case, where no logarithmic oversampling is needed.

The following result extends Theorem 4.7.1 for approximately lower-dimensional distributions.

Theorem 9.2.4 (Covariance estimation for lower-dimensional distributions). *Let X be a sub-gaussian random vector in $\mathbb{R}^n$. More precisely, assume that there exists $K \geq 1$ such that*

$$\|\langle X, x \rangle\|_{\psi_2} \leq K \|\langle X, x \rangle\|_{L^2} \quad \text{for any } x \in \mathbb{R}^n.$$

Then, for every positive integer m , we have

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r}{m}} + \frac{r}{m} \right) \|\Sigma\|.$$

where $r = \text{tr}(\Sigma)/\|\Sigma\|$ is the stable rank of $\Sigma^{1/2}$.

² We introduced the notion of stable rank in Section 7.6.1.

Proof We begin the proof exactly as in Theorem 4.7.1 by bringing the distribution to isotropic position. Thus,

$$\begin{aligned}
\|\Sigma_m - \Sigma\| &= \|\Sigma^{1/2} R_m \Sigma^{1/2}\| \quad \left(\text{where } R_m = \frac{1}{m} \sum_{i=1}^m Z_i Z_i^\top - I_n \right) \\
&= \max_{x \in S^{n-1}} \langle \Sigma^{1/2} R_m \Sigma^{1/2} x, x \rangle \quad (\text{the matrix is symmetric positive semidefinite}) \\
&= \max_{x \in T} \langle R_m x, x \rangle \quad (\text{if we define the ellipsoid } T := \Sigma^{1/2} S^{n-1}) \\
&= \max_{x \in T} \left| \frac{1}{m} \sum_{i=1}^m \langle Z_i, x \rangle^2 - \|x\|_2^2 \right| \quad (\text{by definition of } R_m) \\
&= \frac{1}{m} \max_{x \in T} \left| \|Ax\|_2^2 - m\|x\|_2^2 \right|,
\end{aligned}$$

where in the last step A denotes the $m \times n$ matrix with rows Z_i . As in the proof of Theorem 4.7.1, Z_i are mean zero, isotropic, sub-gaussian random vectors with $\|Z_i\|_{\psi_2} \lesssim 1$. (For simplicity, let us hide the dependence on K in this argument.) This allows us to apply matrix deviation inequality for A (in the form given in Exercise 9.1.10), which gives

$$\mathbb{E} \|\Sigma_m - \Sigma\| \lesssim \frac{1}{m} \left(\gamma(T)^2 + \sqrt{m} \text{rad}(T) \gamma(T) \right).$$

The radius and Gaussian complexity of the ellipsoid $T = \Sigma^{1/2} S^{n-1}$ are easy to compute:

$$\text{rad}(T) = \|\Sigma\|^{1/2} \quad \text{and} \quad \gamma(T) \leq (\text{tr } \Sigma)^{1/2}.$$

(Check!) This gives

$$\mathbb{E} \|\Sigma_m - \Sigma\| \lesssim \frac{1}{m} \left(\text{tr } \Sigma + \sqrt{m \|\Sigma\| \text{tr } \Sigma} \right).$$

Substitute $\text{tr}(\Sigma) = r \|\Sigma\|$ and simplify the bound to complete the proof. $\square$

Exercise 9.2.5 (Tail bound).  Prove a high-probability guarantee for Theorem 9.2.4 (similar to the results of Exercise 4.7.3 and 5.6.4). Namely, check that for any $u \geq 0$, we have

$$\|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r+u}{m}} + \frac{r+u}{m} \right) \|\Sigma\|$$

with probability at least $1 - 2e^{-u}$.

9.3 Johnson-Lindenstrauss Lemma for infinite sets

Let us now apply the matrix deviation inequality for a finite set T . In this case, we recover Johnson-Lindenstrauss Lemma from Section 5.3 and more.

9.3.1 Recovering the classical Johnson-Lindenstrauss

Let us check that matrix deviation inequality contains the classical Johnson-Lindenstrauss Lemma (Theorem 5.3.1). Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$ and define T to be the set of normalized differences of $\mathcal{X}$, i.e.

$$T := \left\{ \frac{x - y}{\|x - y\|_2} : x, y \in \mathcal{X} \text{ are distinct points} \right\}.$$

The Gaussian complexity of T satisfies

$$\gamma(T) \leq C\sqrt{\log N} \quad (9.13)$$

(Recall Exercise 7.5.10). Then matrix deviation inequality (Theorem 9.1.1) implies that the bound

$$\sup_{x, y \in \mathcal{X}} \left| \frac{\|Ax - Ay\|_2}{\|x - y\|_2} - \sqrt{m} \right| \lesssim \sqrt{\log N} \quad (9.14)$$

holds with high probability. To keep the calculation simple, we will be satisfied here with probability 0.99, which can be obtained using Markov's inequality; Exercise 9.1.8 gives better probability. Also, for simplicity we suppressed the dependence on the sub-gaussian norm K .

Multiply both sides of (9.14) by $\frac{1}{\sqrt{m}}\|x - y\|_2$ and rearrange the terms. We obtain that, with high probability, the scaled random matrix

$$Q := \frac{1}{\sqrt{m}}A$$

is an approximate isometry on $\mathcal{X}$, i.e.

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_2 \leq (1 + \varepsilon)\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}.$$

where

$$\varepsilon \lesssim \sqrt{\frac{\log N}{m}}.$$

Equivalently, if we fix $\varepsilon > 0$ and choose the dimension m such that

$$m \gtrsim \varepsilon^{-2} \log N,$$

then with high probability Q is an ε -isometry on $\mathcal{X}$. Thus we recover the classical Johnson-Lindenstrauss Lemma (Theorem 5.3.1).

Exercise 9.3.1. ☕☕ In the argument above, quantify the probability of success and dependence on K . In other words, use matrix deviation inequality to give an alternative solution of Exercise 5.3.3.

9.3.2 Johnson-Lindenstrauss lemma for infinite sets

The argument above does not really depend on $\mathcal{X}$ being a finite set. We only used that $\mathcal{X}$ is finite to bound the Gaussian complexity in (9.13). This means that we

can give a version of Johnson-Lindenstrauss lemma for general, not necessarily finite sets. Let us state such version.

Proposition 9.3.2 (Additive Johnson-Lindenstrauss Lemma). *Consider a set $\mathcal{X} \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then, with high probability (say, 0.99), the scaled matrix*

$$Q := \frac{1}{\sqrt{m}}A$$

satisfies

$$\|x - y\|_2 - \delta \leq \|Qx - Qy\|_2 \leq \|x - y\|_2 + \delta \quad \text{for all } x, y \in \mathcal{X}$$

where

$$\delta = \frac{CK^2w(\mathcal{X})}{\sqrt{m}}$$

and $K = \max_i \|A_i\|_{\psi_2}$.

Proof Choose T to be the difference set, i.e. $T = \mathcal{X} - \mathcal{X}$, and apply matrix deviation inequality (Theorem 9.1.1). It follows that, with high probability,

$$\sup_{x, y \in \mathcal{X}} \left| \|Ax - Ay\|_2 - \sqrt{m}\|x - y\|_2 \right| \leq CK^2\gamma(\mathcal{X} - \mathcal{X}) = 2CK^2w(\mathcal{X}).$$

(In the last step, we used (7.22).) Dividing both sides by $\sqrt{m}$, we complete the proof. $\square$

Note that the error δ in Proposition 9.3.2 is additive, while the classical Johnson-Lindenstrauss Lemma for finite sets (Theorem 5.3.1) has a multiplicative form of error. This may be a small difference, but in general it is necessary:

Exercise 9.3.3 (Additive error).  Suppose a set $\mathcal{X}$ has a non-empty interior. Check that, in order for the conclusion (5.10) of the classical Johnson-Lindenstrauss lemma to hold, one must have $m \geq n$, i.e. no dimension reduction is possible.

Remark 9.3.4 (Stable dimension). The additive version of Johnson-Lindenstrauss Lemma can be naturally stated in terms of the stable dimension of $\mathcal{X}$,

$$d(\mathcal{X}) \asymp \frac{w(\mathcal{X})^2}{\text{diam}(\mathcal{X})^2},$$

which we introduced in Section 7.6. To see this, let us fix $\varepsilon > 0$ and choose the dimension m so that it *exceeds an appropriate multiple of the stable dimension*, namely

$$m \geq (CK^4/\varepsilon^2)d(T).$$

Then in Proposition 9.3.2 we have $\delta \leq \varepsilon \text{diam}(\mathcal{X})$. This means that Q *preserves the distances in $\mathcal{X}$ to within a small fraction of the maximal distance*, which is the diameter of $\mathcal{X}$.

9.4 Random sections: M^* bound and Escape Theorem

Consider a set $T \subset \mathbb{R}^n$ and a random subspace E with given dimension. How large is the typical intersection of T and E ? See Figure 9.2 for illustration. There are two types of answers to this question. In Section 9.4.1 we give a general bound for the expected diameter of $T \cap E$; it is called the M^* bound. The intersection $T \cap E$ can even be empty; this is the content of the *Escape Theorem* which we prove in Section 9.4.2. Both results are consequences of matrix deviation inequality.

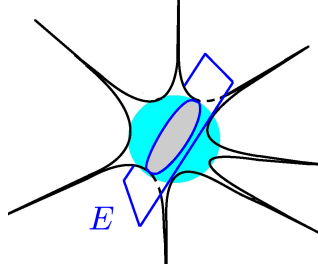


Figure 9.2 Illustration for M^* bound: the intersection of a set T with a random subspace E .

9.4.1 M^* bound

First, it is convenient to realize the random subspace E as a kernel of a random matrix, i.e. set

$$E := \ker A$$

where A is an $m \times n$ random matrix. We always have

$$\dim(E) \geq n - m,$$

and for continuous distributions we have $\dim(E) = n - m$ almost surely.

Example 9.4.1. Suppose A is a *Gaussian matrix*, i.e. has independent $N(0, 1)$ entries. Rotation invariance implies that $E = \ker(A)$ is uniformly distributed in the Grassmanian:

$$E \sim \text{Unif}(G_{n, n-m}).$$

Our main result is the following general bound on the diameters of random sections of geometric sets. For historic reasons, this results is called the M^* bound.

Theorem 9.4.2 (M^* bound). *Consider a set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random subspace $E = \ker A$ satisfies*

$$\mathbb{E} \text{diam}(T \cap E) \leq \frac{CK^2 w(T)}{\sqrt{m}},$$

where $K = \max_i \|A_i\|_{\psi_2}$.

Proof Apply Theorem 9.1.1 for $T - T$ and obtain

$$\mathbb{E} \sup_{x, y \in T} \left| \|Ax - Ay\|_2 - \sqrt{m} \|x - y\|_2 \right| \leq CK^2 \gamma(T - T) = 2CK^2 w(T).$$

If we restrict the supremum to points x, y in the kernel of A , then $\|Ax - Ay\|_2$ disappears since $A(x - y) = 0$, and we have

$$\mathbb{E} \sup_{x, y \in T \cap \ker A} \sqrt{m} \|x - y\|_2 \leq 2CK^2 w(T).$$

Dividing by $\sqrt{m}$ yields

$$\mathbb{E} \text{diam}(T \cap \ker A) \leq \frac{CK^2 w(T)}{\sqrt{m}},$$

which is the bound we claimed. $\square$

Exercise 9.4.3 (Affine sections). ☕☕ Check that M^* bound holds not only for sections through the origin but for all affine sections as well:

$$\mathbb{E} \max_{z \in \mathbb{R}^n} \text{diam}(T \cap E_z) \leq \frac{CK^2 w(T)}{\sqrt{m}}$$

where $E_z = z + \ker A$.

Remark 9.4.4 (Stable dimension). Surprisingly, the random subspace E in the M^* bound is not low-dimensional. On the contrary, $\dim(E) \geq n - m$ and we would typically choose $m \ll n$, so E has almost full dimension. This makes the M^* bound a strong and perhaps surprising statement.

It can be enlightening to look at the M^* bound through the lens of the notion of stable dimension $d(T) \asymp w(T)^2 / \text{diam}(T)^2$, which we introduced in Section 7.6. Fix $\varepsilon > 0$. Then the M^* bound can be stated as

$$\mathbb{E} \text{diam}(T \cap E) \leq \varepsilon \cdot \text{diam}(T)$$

as long as

$$m \geq C(K^4/\varepsilon^2)d(T). \quad (9.15)$$

In words, *the M^* bound becomes non-trivial – the diameter shrinks – as long as the codimension of E exceeds a multiple of the stable dimension of T .*

Equivalently, the dimension condition states that the sum of dimension of E and a multiple of stable dimension of T should be bounded by n . This condition should now make sense from the linear algebraic point of view. For example, if T is a centered Euclidean ball in some subspace $F \subset \mathbb{R}^n$ then a non-trivial bound $\text{diam}(T \cap E) < \text{diam}(T)$ is possible only if

$$\dim E + \dim F \leq n.$$

(Why?)

Let us look at one remarkable example of application of the M^* bound.

Example 9.4.5 (The ℓ_1 ball). Let $T = B_1^n$, the unit ball of the ℓ_1 norm in $\mathbb{R}^n$. Since we proved in (7.18) that $w(T) \asymp \sqrt{\log n}$, the M^* bound (Theorem 9.4.2) gives

$$\mathbb{E} \operatorname{diam}(T \cap E) \lesssim \sqrt{\frac{\log n}{m}}.$$

For example, if $m = 0.1n$ then

$$\mathbb{E} \operatorname{diam}(T \cap E) \lesssim \sqrt{\frac{\log n}{n}}. \quad (9.16)$$

Comparing this with $\operatorname{diam}(T) = 2$, we see that the diameter shrinks by almost $\sqrt{n}$ as a result of intersecting T with the random subspace E that has almost full dimension (namely, $0.9n$).

For an intuitive explanation of this surprising fact, recall from Section 7.5.4 that the “bulk” the octahedron $T = B_1^n$ is formed by the inscribed ball $\frac{1}{\sqrt{n}}B_2^n$. Then it should not be surprising if a random subspace E tends to pass through the bulk and miss the “outliers” that lie closer to the vertices of T . This makes the diameter of $T \cap E$ essentially the same as the size of the bulk, which is $1/\sqrt{n}$.

This example indicates what makes a surprisingly strong and general result like M^* bound possible. Intuitively, the random subspace E tends to pass entirely through the bulk of T , which is usually a Euclidean ball with much smaller diameter than T , see Figure 9.2.

Exercise 9.4.6 (M^* bound with high probability). 🍷🍷 Use the high-probability version of matrix deviation inequality (Exercise 9.1.8) to obtain a high-probability version of the M^* bound.

9.4.2 Escape theorem

In some circumstances, a random subspace E may completely miss a given set T in $\mathbb{R}^n$. This might happen, for example, if T is a subset of the sphere, see Figure 9.3. In this case, the intersection $T \cap E$ is typically empty under essentially the same conditions as in M^* bound.

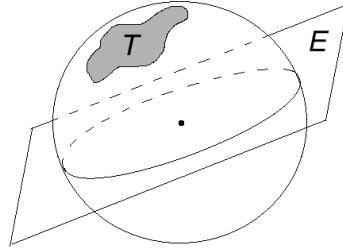


Figure 9.3 Illustration for the Escape theorem: the set T has empty intersection with a random subspace E .

Theorem 9.4.7 (Escape theorem). *Consider a set $T \subset S^{n-1}$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. If*

$$m \geq CK^4 w(T)^2, \quad (9.17)$$

then the random subspace $E = \ker A$ satisfies

$$T \cap E = \emptyset$$

with probability at least $1 - 2 \exp(-cm/K^4)$. Here $K = \max_i \|A_i\|_{\psi_2}$.

Proof Let us use the high-probability version of matrix deviation inequality from Exercise 9.1.8. It states that the bound

$$\sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m} \right| \leq C_1 K^2 (w(T) + u) \quad (9.18)$$

holds with probability at least $1 - 2 \exp(-u^2)$. Suppose this event indeed holds and $T \cap E \neq \emptyset$. Then for any $x \in T \cap E$ we have $\|Ax\|_2 = 0$, so our bound becomes

$$\sqrt{m} \leq C_1 K^2 (w(T) + u).$$

Choosing $u := \sqrt{m}/(2C_1 K^2)$, we simplify the bound to

$$\sqrt{m} \leq C_1 K^2 w(T) + \frac{\sqrt{m}}{2},$$

which yields

$$\sqrt{m} \leq 2C_1 K^2 w(T).$$

But this contradicts the assumption of the Escape theorem, as long as we choose the absolute constant C large enough. This means that the event (9.18) with u chosen as above implies that $T \cap E = \emptyset$. The proof is complete. $\square$

Exercise 9.4.8 (Sharpness of Escape theorem). $\clubsuit$ Discuss the sharpness of Escape Theorem for the example where T is the unit sphere of some subspace of $\mathbb{R}^n$.

Exercise 9.4.9 (Escape from a point set). $\clubsuit\clubsuit$ Prove the following version of Escape theorem with a rotation of a point set instead of a random subspace.

Consider a set $T \subset S^{n-1}$ and let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$. Show that, if

$$\sigma_{n-1}(T) < \frac{1}{N}$$

then there exists a rotation $U \in O(n)$ such that

$$T \cap U\mathcal{X} = \emptyset.$$

Here σ_{n-1} denotes the normalized Lebesgue measure (area) on S^{n-1} .

Hint: Consider a random rotation $U \in \text{Unif}(SO(n))$ as in Section 5.2.5. Applying a union bound, show that the probability that there exists $x \in \mathcal{X}$ such that $Ux \in T$ is smaller than 1.

9.5 Notes

Matrix deviation inequality (Theorem 9.1.1) and its proof are borrowed from [132]. Several important related results had been known before. In the partial case where A is Gaussian and T is a subset of the unit sphere, Theorem 9.1.1, can be deduced from Gaussian comparison inequalities. The upper bound on $\|Gx\|_2$ can be derived from Sudakov-Fernique’s inequality (Theorem 7.2.11), while the lower bound can be obtained from Gordon’s inequality (Exercise 7.2.14). G. Schechtman [181] proved a version of matrix deviation inequality in the partial case of Gaussian random matrices A and for general norms (not necessarily Euclidean); we present this result in Section 11.1. For sub-gaussian matrices A , some earlier versions of matrix deviation inequality can be found in [117, 145, 63]; see [132, Section 3] for comparison with these results. Finally, a variant of matrix deviation inequality for sparse matrices A (more precisely, for A being the sparse Johnson-Lindenstrauss transform) is obtained in [31].

The quadratic dependence on K in Theorem 9.1.1 was improved to $K\sqrt{\log K}$ in [109], which is optimal. This automatically leads to corresponding improvements of the dependence on subgaussian norms in all results that can be obtained as consequences of Theorem 9.1.1, including Theorem 3.1.1, Theorem 4.6.1, Proposition 9.2.1, Theorem 5.3.1, Theorem 9.4.2, Theorem 9.4.7, Theorem 10.2.1, Corollary 10.3.4, etc.

A version of Proposition 9.2.1 is due to V. Milman [149]; see [11, Proposition 5.7.1]. Theorem 9.2.4 on covariance estimation for lower-dimensional distributions is due to V. Koltchinskii and K. Lounici [119]; they used a different approach that was also based on the majorizing measure theorem. R. van Handel shows in [213] how to derive Theorem 9.2.4 for Gaussian distributions from decoupling, conditioning and Slepian Lemma. The bound in Theorem 9.2.4 can be reversed [119, 213].

A version of Johnson-Lindenstrauss lemma for infinite sets similar to Proposition 9.3.2 is from [132].

The M^* bound, a version of which we proved in Section 9.4.1, is an useful result in geometric functional analysis, see [11, Section 7.3–7.4, 9.3], [87, 144, 223] for many known variants, proofs and consequences of M^* bounds. The version we gave here, Theorem 9.4.2, is from [132].

The escape theorem from Section 9.4.2 is also called “escape from the mesh” in the literature. It was originally proved by Y. Gordon [87] for a Gaussian random matrix A and with a sharp constant factor in (9.17). The argument was based on Gordon’s inequality from Exercise 7.2.14. Matching lower bounds for this sharp theorem are known for spherically convex sets [190, 9]. In fact, for a spherically convex set, the exact value of the hitting probability can be obtained by the methods of integral geometry [9]. Oymak and Tropp [163] proved that this sharp result is universal, i.e. can be extended for non-Gaussian matrices. Our version of escape theorem (Theorem 9.4.7), which is valid for even more general class of random matrices but does not feature sharp absolute constants, is borrowed from

[132]. As we will see in Section 10.5.1, the escape theorem is an important tool for signal recovery problems.